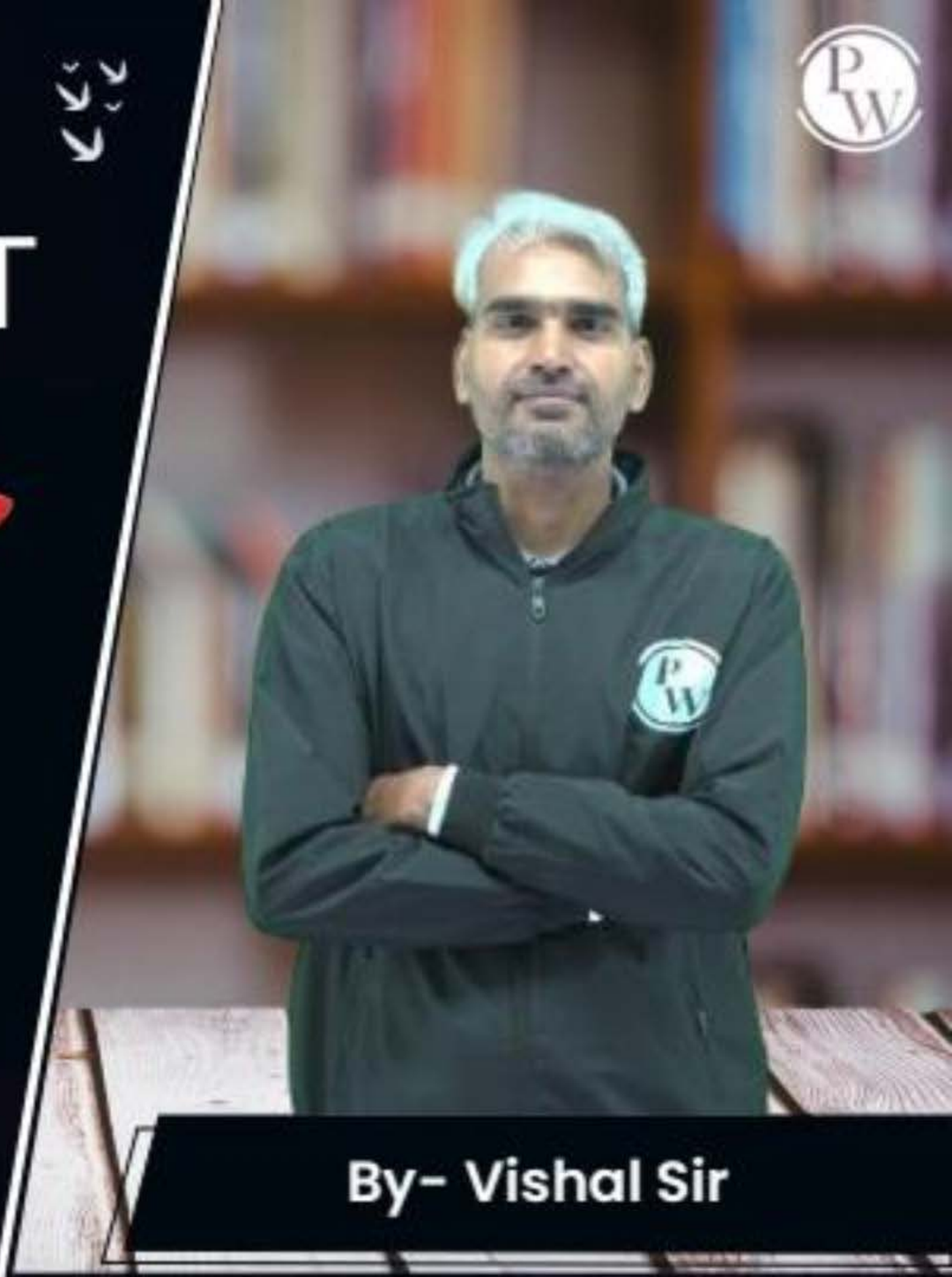
Computer Science & IT



Database Management System

Transactions and Concurrency control

Lecture No. 04



By- Vishal Sir

Recap of Previous Lecture



- Topic: Equivalent schedule
- Topic: Serializable schedule
- Topic: RW problem
- Topic: WR problem
- Topic: WW problem
- Topic: Lost update problem

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Topics to be Covered



- Topic: Classification based on recoverability
- Topic: Irrecoverable schedule
- Topic: Recoverable schedule
- Topic: Cascading rollback problem
- Topic: Cascadeless recoverable schedule
- Topic: Strict recoverable schedule

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Topic : Classification of schedule

Classification of Schedules

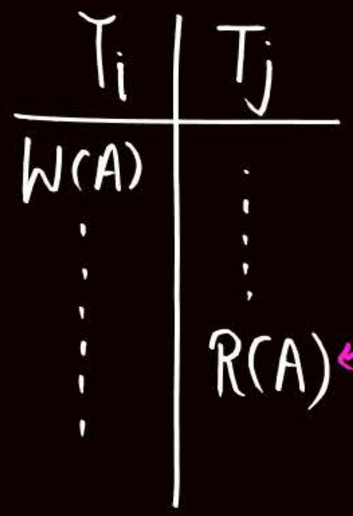
Based on Recoverability

1. Irrecoverable Schedule
2. Recoverable Schedule
3. Cascadeless recoverable Schedule
4. Strict recoverable schedule

Based on Serializability

1. Conflict Serializable Schedule
2. View Serializable Schedule

* If transaction Tj reads the value updated by an uncommitted transaction Ti then we say that transaction Tj is dependent on transaction Ti.
  * In this case if we rollback the transaction Ti then we must also rollback the transactions dependent on transaction Ti {eg: Tj}

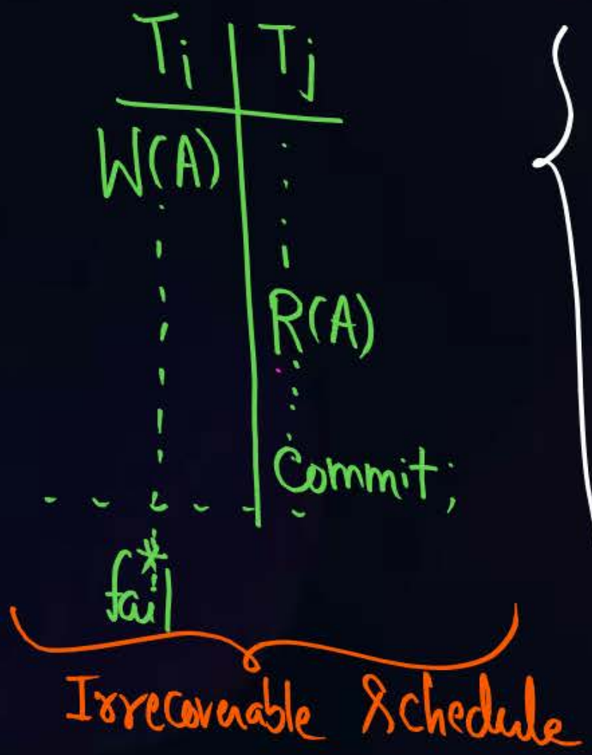


Dirty Read (or) Uncommitted Read



Topic : Irrecoverable schedule

* Rollbacking of Committed transaction is not possible,
  ∴ If there is a situation when we have to rollback a Committed transaction then it is not possible and Schedule becomes irrecoverable Schedule.



Tj is dependent on transaction Ti.
When transaction Ti fails then we rollback transaction Ti and because of rollbacking of transaction Ti we will try to rollback the transaction Tj, but we can not rollback transaction Tj because Tj has Committed
∴ Schedule becomes irrecoverable Schedule

eg:

T_1	T_2
W(A)	
	R(A)
	Commit ;
Commit	

T_1 can fail anywhere in between

it is unsafe

→ After Commit of T_2, we can not be sure that transaction T_1 will not fail and Commit.

∴ It is unsafe

Schedule is irrecoverable schedule

eg:

T_1	T_2
W(A)	
	R(A)

No one has performed Commit opn

∴ Both can rollback

Hence, Recoverable Schedule

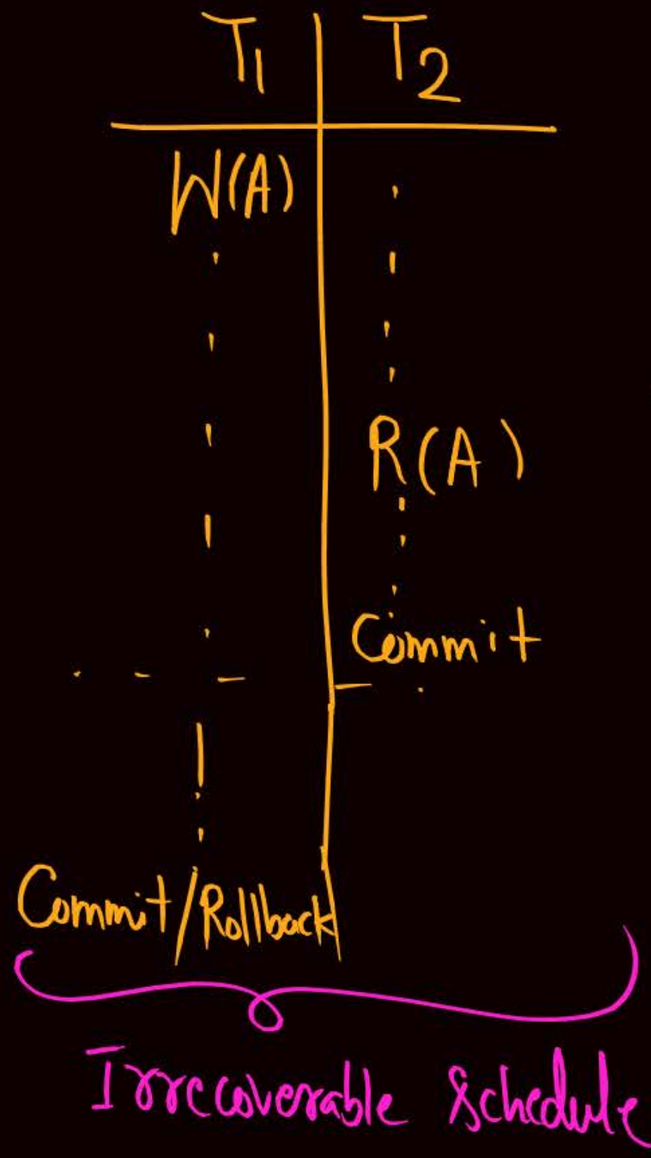
eg:

T_1	T_2
W(A)	
Rollback/Commit	R(A)

Because T_2 has not Committed

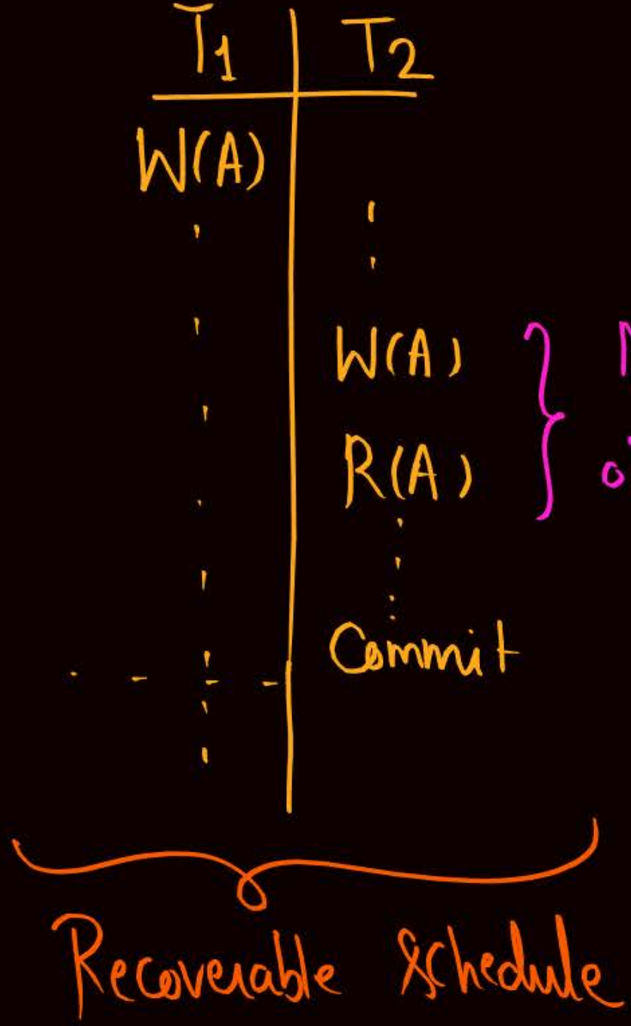
∴ we can rollback T_2 as well if needed

Hence, Recoverable Schedule



If transaction T_2 reads the value updated by an uncommitted transaction T_1 and transaction T_2 commits before the commit or rollback of transaction T_1 then schedule is called Irrecoverable schedule

eg:



No uncommitted read op^n
$\therefore$ T_2 does not depend on T_1

Hence, we don't need to rollback transaction T_2 even if transaction T_1 fails

$\therefore$ Recoverable schedule

Recoverable schedule

* Irrecoverability may be possible only if uncommitted read oph exists.

*

If there is uncommitted read oph in the Schedule, then Schedule will always be recoverable



Topic : Recoverable schedule

For a schedule to be recoverable,

Either no uncommitted read op^n should exist,

Or if any uncommitted read op^n exists in the schedule then the transaction with uncommitted read op^n must not Commit until the transaction on which it is dependent has performed Commit or Rollback op^n.

T_1	T_2
W(A)	
	R(A)
C_1/R_1	
	Commit

Recoverable



Topic : Cascading rollback problem

* If failure of one transaction results in rollbacking of some other transactions as well along with the rollbacking of failed transaction then it is called "Cascading rollback" problem.



Topic : Cascading rollback problem

Problem because of Cascading rollback is wastage of CPU time and wastage of IO time



T1 fails

↳ Rollback T_1 and also rollback the transactions dependent on T_1 {i.e. T_2 & T_3}

* Because of Rollbacking of T_3 rollback the transactions dependent on T_3 {i.e, T_4 & T_5}
* Because of rollbacking of T_5 we will have to rollback T_6



Topic : Cascading rollback problem

- If uncommitted read opn exists in the schedule, then dependency will exist among the transaction, and because of that Cascading rollback problem will exist in the schedule.



Topic : Cascadeless recoverable schedule

* For cascadeless recoverable schedule these should be no dependency among the transactions. i.e., Uncommitted read oph must not exist in the schedule

* To ensure cascadeless recoverable schedule, uncommitted oph must not be allowed.

T_1	T_2
W(A)	:
	R(B) ✓ Allowed
Commit	:
	R(A)

i.e. If transaction T_i has updated dataitem A, then no other transaction should be allowed to Read the dataitem 'A' until the Commit/Rollback of T_i.



Topic : Cascadeless recoverable schedule

In Cascadeless recoverable schedule uncommitted Read opn not allowed

∴

T_1	T_2
W(A)	
Commit/Rollback	
	R(A)

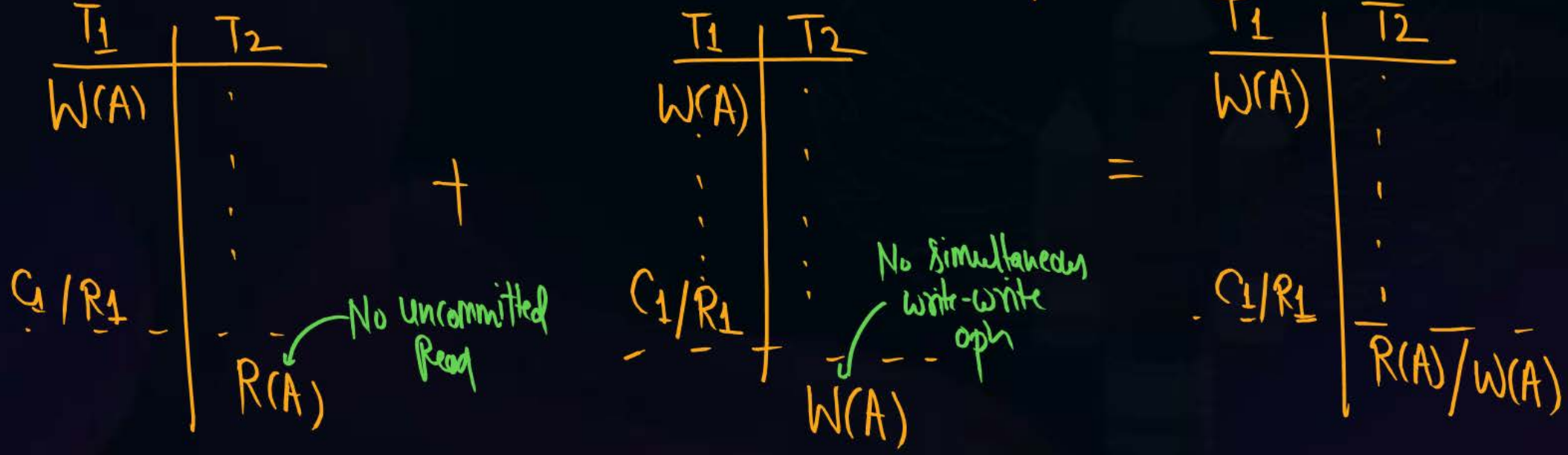
Cascadeless Recoverable Schedule

- free from
  - ① Cascading problem
  - ② WR problem
- Not free from
  - ① RW problem
  - ② WW problem
  - ③ lost update problem

Topic : Strict recoverable schedule

Strict recoverable schedule is free from cascading roll back problem and free from lost update problem.

Schedule free from Cascading Rollback + Schedule free from lost update problem (No simultaneous write-write opn) = Strict Recoverable Schedule





2 mins Summary

- **Topic** Classification based on recoverability
- **Topic** Irrecoverable schedule
- **Topic** Recoverable schedule
- **Topic** Cascading rollback problem
- **Topic** Cascadeless recoverable schedule
- **Topic** Strict recoverable schedule

THANK - YOU

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